2021

一. 1. (a) $f(z)=\frac{1}{(z-1)(z-3)}=\frac{1}{2}\left(\frac{1}{z-3}-\frac{1}{z-1}\right)$ $2<|z-1|<+\infty$

$$f(z)=\frac{1}{2}\left[\frac{1}{(z-1)\left(1-\frac{2}{z-1}\right)}-\frac{1}{z-1}\right]$$

$$=\frac{1}{2}\left[\frac{1}{z-1}\sum_{k=0}^{\infty}\left(\frac{2}{z-1}\right)^k-\frac{1}{z-1}\right]=\sum_{k=1}^{\infty}\frac{2^{k-1}}{(z-1)^{k+1}}$$

(b) $f(z)=\frac{1}{z^2(z-i)}$ $0<|z-i|<1$

$$\frac{1}{z}=\frac{1}{i}\cdot\frac{1}{1-i(z-i)}=\sum_{k=0}^{\infty}i^{k-1}(z-i)^k$$

$$\frac{1}{z^2}=\frac{d}{dz}\left(-\frac{1}{z}\right)=-\sum_{k=0}^{\infty}[i(z-i)]^{k-1}\cdot k=-\sum_{k=1}^{\infty}k\,i^{k-1}(z-i)^{k-1}$$

$$\frac{1}{z^2(z-i)}=-\sum_{k=1}^{\infty}k\,i^{k-1}(z-i)^{k-2}=-\sum_{k=0}^{\infty}(k+1)\,i^k(z-i)^{k-1}$$

2. (a) $f(z)=\frac{1}{z^2(1-3z^2)}$

奇点：$z_1=0$ (二阶极点)

$z_2=\sqrt{\frac{1}{3}}$ 单极点 $z_3=-\sqrt{\frac{1}{3}}$ 单极点

$$\mathrm{Res}\,f(z_1)=\lim_{z\to0}\frac{d}{dz}\left(z^2\cdot\frac{1}{z^2(1-3z^2)}\right)=\lim_{z\to0}\frac{d}{dz}\frac{1}{1-3z^2}=0$$

$$\mathrm{Res}\,f(z_2)=\lim_{z\to\sqrt{\frac{1}{3}}}\left(z-\sqrt{\tfrac{1}{3}}\right)\frac{1}{-3z^2(z+\sqrt{\frac{1}{3}})(z-\sqrt{\frac{1}{3}})}=\frac{1}{-2\sqrt{3}}=-\frac{\sqrt{3}}{2}$$

$$\mathrm{Res}\,f(z_3)=\frac{\sqrt{3}}{2}$$

(b) $f(z)=\frac{z^2e^{1/z}}{z-1}$ 奇点：$z_1=1$ (单极点) $z_2=0$ 本性极点

$\mathrm{Res}\,f(z_1)=e$ $\mathrm{Res}\,f(z_2)=a_{-1}$

$$\frac{z^2e^{1/z}}{z-1}=-z^2\left(1+z^{-1}+\frac{1}{2!}z^{-2}+\cdots\right)(1+z+z^2+\cdots)\quad -a_{-1}=\frac{1}{3!}+\frac{1}{4!}+\cdots$$

$$\therefore\mathrm{Res}\,f(z_2)=-\sum_{k=3}^{\infty}\frac{1}{k!}$$

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3. (a) $\int_0^{2\pi} \tan(\theta+\frac{i}{2})\,d\theta = \int_0^{2\pi} \frac{\sin(\theta+\frac{i}{2})}{\cos(\theta+\frac{i}{2})}\,d\theta$

令 $z = e^{i(\theta+\frac{i}{2})} = e^{i\theta-\frac{1}{2}}$ $\quad d\theta = \frac{dz}{iz}$

$\Rightarrow \int_0^{2\pi} \frac{\sin(\theta+\frac{i}{2})}{\cos(\theta+\frac{i}{2})} = \oint_{|z|=e^{-\frac{1}{2}}} \frac{\frac{1}{2i}(z-z^{-1})}{\frac{1}{2}(z+z^{-1})} \frac{dz}{iz} = \oint_{|z|=e^{-\frac{1}{2}}} -\frac{z^2-1}{z(z^2+1)}$

$z=0$ 在 $|z|=e^{-\frac{1}{2}}$ 内部 $\therefore \oint_{|z|=e^{-\frac{1}{2}}} -\frac{z^2-1}{z(z^2+1)} = -2\pi i \left(\frac{z^2-1}{z^2+1}\right)\Big|_{z=0} = 2\pi i$

(b) $\int_0^\infty \frac{\sin 3x}{x(x^2+4)}\,dx$ $\quad h(z) = \frac{1}{z(z^2+4)}$ $\quad \alpha=0,\ z_1=2i,\ z_2=-2i$

$\because \int_0^\infty h(x)\sin mx\,dx = \pi \sum \mathrm{Res}\left(h(z)e^{imz}\right) + \frac{\pi}{2}\sum \mathrm{Res}\left(h(\alpha)e^{im\alpha}\right)$

$\therefore \int_0^\infty \frac{\sin 3x}{x(x^2+4)} = \pi\,\mathrm{Res}\left(h(z_1)e^{3iz_1}\right) + \frac{\pi}{2}\mathrm{Res}\left(h(\alpha)e^{3i\alpha}\right)$

$= \pi \cdot \frac{e^{-6}}{-8} + \frac{\pi}{2}\cdot\frac{1}{4} = \frac{\pi}{8}(1-e^{-6})$

4. (a) $\mathcal{F}[\sin(2x)\,H(x-1)] = \mathcal{F}[(e^{i2x}-e^{-i2x})\,H(x-1)]/2i$

$= \frac{1}{2i}\left(\mathcal{F}[e^{2ix}H(x-1)] - \mathcal{F}[e^{-2ix}H(x-1)]\right) = \frac{1}{2i}\left(\hat{g}(\underline{w-2}) - \hat{g}(\underline{w+2})\right)$

$\hat{g}(w) = \mathcal{F}[H(x-1)] = e^{-iw}\mathcal{F}[H(x)] = e^{-iw}\left(\frac{1}{iw\sqrt{2\pi}} + \sqrt{\frac{\pi}{2}}\delta(w)\right)$

$\therefore = \frac{e^{-iw}}{2i}\left[\frac{1}{i\sqrt{2\pi}(w-2)} + \sqrt{\frac{\pi}{2}}\delta(w-2) - \frac{1}{i\sqrt{2\pi}(w+2)} - \sqrt{\frac{\pi}{2}}\delta(w+2)\right)$

(b) $\mathcal{L}[te^t\cos 2t] = -\frac{d}{dp}\mathcal{L}[e^t\cos 2t]$

$\mathcal{L}[e^t\cos 2t] = \int_0^\infty e^t\cos 2t\,e^{-pt}\,dt = \int_0^\infty e^{-(p+1)t}\frac{e^{2it}+e^{-2it}}{2}\,dt$

$= \frac{1}{2}\int_0^\infty \left(e^{(-p-1+2i)t} + e^{(-p-1-2i)t}\right)dt = \frac{1}{2}\left(\frac{1}{p+1-2i} + \frac{1}{p+1+2i}\right) = \frac{p+1}{(p+1)^2+4}$

$\therefore$ 原 $= -\frac{d}{dp}\frac{p+1}{(p+1)^2+4} = -\frac{(p+1)^2+4-2(p+1)^2}{((p+1)^2+4)^2} = \frac{(p+1)^2-4}{((p+1)^2+4)^2}$

(c) $f(x) = \begin{cases} e^{-x} & x \geq 0 \\ 0 & x < 0 \end{cases}$

$$f * f = \int_{-\infty}^{\infty} f(\xi)\, f(x-\xi)\, d\xi = \begin{cases} x \leq 0 : 0 \\ x > 0 : = \int_0^x e^{-\xi} \cdot e^{-(x-\xi)}\, d\xi = x e^{-x} \end{cases}$$

$$f * f = \begin{cases} x e^{-x} & x > 0 \\ 0 & x \leq 0 \end{cases}$$

$=$.

1. $\partial_t u - a^2 \Delta u = 0 \qquad u = (t, x, y, z) \to (t, r, \theta, \varphi)$

$$\begin{cases} x = r \sin\theta \cos\varphi \\ y = r \sin\theta \sin\varphi \\ z = r \cos\theta \end{cases} \qquad J = \begin{pmatrix} \partial_r x & \partial_\theta x & \partial_\varphi x \\ \partial_r y & \partial_\theta y & \partial_\varphi y \\ \partial_r z & \partial_\theta z & \partial_\varphi z \end{pmatrix} = \begin{pmatrix} \sin\theta\cos\varphi & r\cos\theta\cos\varphi & -r\sin\theta\sin\varphi \\ \sin\theta\sin\varphi & r\cos\theta\sin\varphi & r\sin\theta\cos\varphi \\ \cos\theta & -r\sin\theta & 0 \end{pmatrix}$$

$$J^{-1} = \begin{pmatrix} \frac{\partial r}{\partial x} & \frac{\partial \theta}{\partial x} & \frac{\partial \varphi}{\partial x} \\ \frac{\partial r}{\partial y} & \frac{\partial \theta}{\partial y} & \frac{\partial \varphi}{\partial y} \\ \frac{\partial r}{\partial z} & \frac{\partial \theta}{\partial z} & \frac{\partial \varphi}{\partial z} \end{pmatrix} = \begin{pmatrix} \sin\theta\cos\varphi & \sin\theta\sin\varphi & \cos\theta \\ \frac{1}{r}\cos\theta\cos\varphi & \frac{1}{r}\cos\theta\sin\varphi & -\frac{1}{r}\sin\theta \\ -\frac{1}{r\sin\theta}\sin\varphi & \frac{1}{r\sin\theta}\cos\varphi & 0 \end{pmatrix}$$

$$\frac{\partial r}{\partial x} \sin\theta\cos\varphi = \frac{\partial r}{\partial x}\frac{\partial x}{\partial r} + \underbrace{\frac{\partial \theta}{\partial x}\frac{\partial x}{\partial \theta}}_{=0} + \underbrace{\frac{\partial \varphi}{\partial x}\frac{\partial x}{\partial \varphi}}_{=0} = \frac{\partial x}{\partial r}$$

- $\frac{\partial^2 u}{\partial x^2} = \sin^2\theta\cos^2\varphi \frac{\partial^2 u}{\partial r^2} + \cos\varphi \frac{\partial u}{\partial r}\frac{\partial \theta}{\partial x}\cos\theta - \sin\theta \frac{\partial u}{\partial r}\frac{\partial \varphi}{\partial x}\sin\varphi$

$= \sin^2\theta\cos^2\varphi \frac{\partial^2 u}{\partial r^2} + \frac{1}{r}(\cos^2\theta\cos^2\varphi + \sin^2\varphi)\frac{\partial u}{\partial r}$

- $\frac{\partial^2 u}{\partial y^2} = \sin^2\theta\sin^2\varphi \frac{\partial^2 u}{\partial r^2} + \frac{1}{r}(\cos^2\theta\sin^2\varphi + \cos^2\varphi)\frac{\partial u}{\partial r}$

- $\frac{\partial^2 u}{\partial z^2} = \cos^2\theta \frac{\partial^2 u}{\partial r^2} + \frac{1}{r}\sin^2\theta \frac{\partial u}{\partial r}$

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$$\Rightarrow \Delta u = \frac{\partial^2 u}{\partial r^2} + \frac{2}{r}\frac{\partial u}{\partial r}$$

$$\therefore \text{原} \Leftrightarrow \partial_t u - a^2\left(\frac{\partial^2 u}{\partial r^2} + \frac{2}{r}\frac{\partial u}{\partial r}\right) = 0$$

$$\Leftrightarrow \frac{\partial (ru)}{\partial t} - a^2\left(r\frac{\partial^2 u}{\partial r^2} + 2\frac{\partial u}{\partial r}\right) = 0$$

$$\Leftrightarrow \frac{\partial (ru)}{\partial t} - a^2 \frac{\partial^2 (ru)}{\partial r^2} = 0$$

(2) $\begin{cases} \partial_{tt} u - \partial_{xx} u = 0 \\ u(x,0) = 0, \quad \partial_t u(x,0) = \psi(x) \end{cases}$

$\xrightarrow{\text{Fourier}}$ $\begin{cases} \partial_{tt}\hat{u} + w^2 \hat{u} = 0 \\ \hat{u}(w,0) = 0, \quad \partial_t \hat{u}(w,0) = \hat{\psi}(w) \end{cases}$

$\hat{u} = c_1 e^{iwt} + c_2 e^{-iwt}$ $\quad \begin{cases} c_1 + c_2 = 0 \\ 2iw c_1 = \hat{\psi} \end{cases}$ $\Rightarrow \hat{u} = \hat{\psi}\frac{\sin wt}{w}$

$\Rightarrow u(x,t) = \frac{1}{\sqrt{2\pi}} \psi * \mathcal{F}^{-1}\left[\frac{\sin wt}{w}\right] = \frac{1}{2}\int_{x-t}^{x+t} \psi$

(3) $\begin{cases} \partial_t u - \partial_{xx} u = 0 & 0<x<1,\ t>0 \\ \partial_x u(0,t) = \partial_x u(1,t) + 2u(1,t) = 0 & t>0 \\ u(x,0) = \cos \pi x & 0 \le x \le 1 \end{cases}$ $\quad u_k \tan u_k = 2$

$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X'(0) = 0, \quad X'(1) + 2X(1) = 0 \end{cases}$ $\Rightarrow X(x) = \cancel{C_1 \sin\sqrt{\lambda}x} + C_2 \cos\sqrt{\lambda}x$

$-\sqrt{\lambda}\sin\sqrt{\lambda} + 2\cos\sqrt{\lambda} = 0 \Rightarrow \sqrt{\lambda}\tan\sqrt{\lambda} = 2$

$\lambda_k = u_k^2, \quad X_k(x) = \cos u_k x \quad \Rightarrow T_k'(t) + \lambda_k T_k(t) = 0 \Rightarrow T_k(t) = e^{-\lambda_k t} = e^{-u_k^2 t}$

$$u(x,t) = \sum_{k=1}^{\infty} A_k e^{-u_k^2 t} \cos u_k x$$

$$\text{Camp} A_k = \frac{\int_0^1 \cos \pi\xi \cos u_k \xi \, d\xi}{\int_0^1 \cos^2 u_k \xi \, d\xi}$$

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(4) $\frac{\oplus}{\ominus}$ $\frac{1}{2\pi}\left(\ln\frac{1}{\sqrt{(x-\xi)^2+(y-\eta)^2}}-\ln\frac{1}{\sqrt{(x+\xi)^2+(y-\eta)^2}}\right)$

三. $\begin{cases} \partial_t u - a^2(\partial_{rr}u+\frac{1}{r}\partial_r u)=0 & 0<r<\frac{1}{2},\ t>0 \\ \partial_r u(\frac{1}{2},t)=0,\ |u(0,t)|<\infty & t>0 \\ u(r,0)=1-4r^2 & 0\le r\le\frac{1}{2} \end{cases}$

(1) $u(r,t)=R(r)T(t)$ 代入.

$\begin{cases} RT'-a^2(R''T+\frac{1}{r}R'T)=0 \\ R'(\frac{1}{2})T(t)=0,\ |R(0)T(t)|<\infty \end{cases}$ 设 $\frac{T'}{a^2T}=\frac{R''+\frac{1}{r}R'}{R}=-\lambda$

$\Rightarrow \begin{cases} R''+\frac{1}{r}R'+\lambda R=0 \\ R'(\frac{1}{2})=0,\ |R(0)|<\infty \\ T'+a^2\lambda T=0 \end{cases}$ $\xrightarrow{}$ $\int_0^{\frac{1}{2}} rRR''+RR'+\lambda R^2 r\,dr=0$

$=0-\int_0^{\frac{1}{2}}(rR)'R'\,dr+\int_0^{\frac{1}{2}}RR'\,dr$

$\Rightarrow \lambda\int_0^{\frac{1}{2}}R^2 r\,dr=\int_0^{\frac{1}{2}} r(R')^2\,dr \quad \Rightarrow \lambda\ge 0$

(2)

① $\lambda_0=0$ $R(r)=1$

② $\lambda>0$ $R''+\frac{1}{r}R'+\lambda R=0$ 为零阶 Bessel $\therefore R(r)=AJ_0(\sqrt{\lambda}r)+BY_0(\sqrt{\lambda}r)$

$\because Y_0(0)=\infty \therefore B=0$

$\Rightarrow R(r)=AJ_0(\sqrt{\lambda}r)$ 又 $\because R'(\frac{1}{2})=0 \therefore J_1(\frac{\sqrt{\lambda}}{2})=0$

设 χ_k 为 J_1 非负零点 $\therefore \lambda_k=4\chi_k^2$ $R_k(r)=J_0(2\chi_k r)$

综合①②, $\lambda_k=4\chi_k^2$ $k=0,1,2,\cdots$

$R_k(r)=J_0(2\chi_k r)$ $k=0,1,2,\cdots$

(3) $\int_0^{\frac{1}{2}} r J_0(2\chi_k r)J_0(2\chi_l r)\,dr=\begin{cases} 0 & k\neq l \quad \text{—— 正交性} \\ \frac{1}{8}J_0^2(\chi_k) & k=l \quad \text{—— 平方值} \end{cases}$

(4) λ_k 代入 $T'+a^2\lambda T=0 \Rightarrow T=A_k e^{-4a^2\chi_k^2 t}$ $\Rightarrow u(r,t)=\sum_{k=0}^{\infty}A_k e^{-4a^2\chi_k^2 t}J_0(2\chi_k r)$

$A_k=\frac{\int_0^{\frac{1}{2}}(1-4r^2)J_0(2\chi_k r)\,dr}{\int_0^{\frac{1}{2}} rJ_0^2(2\chi_k r)\,dr}=\begin{cases} \frac{3}{4} & k=0 \\ \frac{-4}{\chi_k^2 J_0(\chi_k)} & k\neq 0 \end{cases}$

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$k \neq 0$

① $\int_0^{\frac{1}{2}} r J_0(2x_k r)\,dr = \int_0^{\frac{1}{2}} (2x_k r) J_0(2x_k r)\, d(2x_k r) \;/\; 4x_k^2$

$= \frac{1}{4x_k^2} \int_0^{x_k} x J_0(x)\,dx = \frac{1}{4x_k^2} \int_0^{x_k} (x J_1)'\,dx = \frac{J_1(x_k)}{4x_k} = 0$

② $\int_0^{\frac{1}{2}} r^3 J_0(2x_k r)\,dr = \int_0^{\frac{1}{2}} (2x_k r)^3 J_0(2x_k r)\, d(2x_k r) \;/\; 16x_k^4$

$= \frac{1}{16x_k^4} \int_0^{x_k} x^3 J_0(x)\,dx = \frac{1}{16x_k^4} \int_0^{x_k} x^2 (x J_1)'\,dx = \frac{1}{16x_k^4} \int_0^{x_k} x^2 J_1\,dx$

$= -\frac{1}{8x_k^4} \int_0^{x_k} (x^2 J_2)'\,dx = -\frac{J_2(x_k)}{8x_k^2} = -\frac{1}{8x_k^2}\left(\frac{J_1(x_k)}{x_k} - J_0(x_k)\right) = \frac{J_0(x_k)}{8x_k^2}$

③ $\int_0^{\frac{1}{2}} r J_0^2(2x_k r)\,dr = \int_0^{\frac{1}{2}} 2x_k r J_0^2(x_k r)\, d(x_k r) \;/\; 4x_k^2$

$= \frac{1}{4x_k^2} \int_0^{x_k} x J_0^2(x)\,dx = \frac{1}{4x_k^2} \int_0^{x_k} (x J_1(x))' J_0(x)\,dx = \frac{1}{4x_k^2} \int_0^{x_k} x J_1 J_0'\,dx$

③ $= \frac{1}{8} J_0^2(x_k)$

$\therefore A_k = \frac{-\frac{1}{2} J_0(x_k)/x_k^2}{\frac{1}{8} J_0^2(x_k)} = \frac{-4}{x_k^2 J_0(x_k)}$ $(k \neq 0)$

$k=0$ 分子 $= \int_0^{\frac{1}{2}} (r - 4r^3)\,dr = \frac{1}{2}(r^2 - r^4)\Big|_0^{\frac{1}{2}} = \frac{3}{32}$

分母 $= \frac{1}{8}$

$\therefore \frac{3}{4}$

(3) 记 $R_1(r) = J_0(2x_k r)$ $R_2(r) = J_0(2x_l r)$ s.t. $R'' + \frac{1}{r}R' + 4x_k^2 R = 0$

$\Rightarrow \begin{cases} \frac{d}{dr}(rR_1') + 4x_k^2 r R_1 = 0 & (i) \\ \frac{d}{dr}(rR_2') + 4x_l^2 r R_2 = 0 & (ii) \end{cases} \xrightarrow[\int_0^{\frac{1}{2}}]{(i)\cdot R_2 - (ii)\cdot R_1} \int_0^{\frac{1}{2}} \left[R_2 (rR_1')' - R_1 (rR_2')'\right] dr + 4(x_k^2 - x_l^2) \Big|_0^{\frac{1}{2}} r R_1 R_2\,dr = 0$

$\Rightarrow \int_0^{\frac{1}{2}} r R_1 R_2\,dr = -\frac{J_0(2x_k r_0) x_k J_1'(x_k)}{4(x_k^2 - x_l^2)} = \begin{cases} 0 & k \neq l \\ \frac{1}{8} J_0^2(x_k) & k = l \end{cases}$